

On the Extinction Ratio penalty and Sensitivity of next-generation 120 Gb/s upstream IM solutions for Very High-Speed PONs

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Abstract *The sensitivity and impact of finite extinction ratio on proposed IM upstream solutions for next generation 100-120 Gb/s VHSP is investigated* ©2026 The Author(s)

Introduction

The International Telecommunication Union's Telecommunication Standardization Sector (ITU-T) recently started a supplement project G.sup88 on very high-speed PONs (VHSP) that can operate at rates above 50 Gb/s per wavelength [1]. While coherent solutions are considered for VHSP, it is commonly accepted that to keep cost low, intensity modulation (IM) with direct detection (DD) is the preferred technology, like previous generations of PONs [2]. One candidate technology for VHSP is a 120 Gb/s/λ IM-DD system. Due to the reduced dispersion tolerance at 120 Gb/s this will limit IM-DD systems to operate close to the zero-dispersion wavelength [3-4]. Since coexistence with GPON is in scope for G.sup88 and is occupying 1290-1330 nm for the upstream it will make enabling IM-DD systems for VHSP challenging. Therefore G. sup88 is currently investigating so called hybrid solutions. ONU cost is most critical, so the aim is to still enable DD at the ONU for the downstream as well as IM for the upstream. To increase the dispersion tolerance for IM, a coherent receiver is proposed at OLT side (which can be more premium) since it is well known that coherent detection allows for full mitigation of the dispersion penalty in DSP. This will enable IM transmission in the C-band for the upstream, conceivably making coexistence with previous PON generations effortless. Another advantage of coherent detection is the stated superior receiver sensitivity compared to DD.

In this paper we will investigate different upstream IM-based solutions for VHSP. We will measure and compare sensitivities of NRZ transmitted signals, detected by either an optical pre-amplified DD or a coherent (COH) receiver. Since at ONU side a transmitter with a limited extinction ratio (ER) is commonly used, we will investigate sensitivities for different ER and provide simplified equations for the ER penalties [5-6] which are fitted to the measurements, which has not been reported in such detail before. Furthermore, we will focus on using low complexity-based DSP as much as possible to ease the requirements for the upstream under burst-mode (BM) operation.

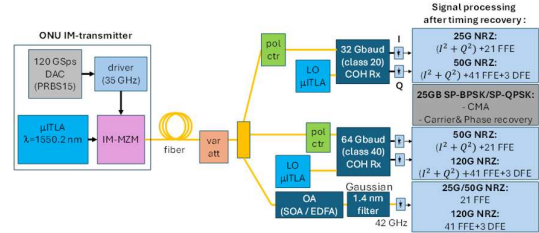


Fig. 1: Experimental setup used to study next generation IM upstream solutions for 100-120 Gb/s/λ PONs

Moreover, we will show that by using DD based DSP we can enable bandwidth limited COH detection of IM signals enabling lower bandwidth coherent receivers and thereby significantly improving cost-effectiveness.

Experimental Setup

Figure 1 depicts the experimental setup used. PRBS15 patterns were generated and uploaded to a 120 GS/s DAC. The NRZ signals were amplified by a 35 GHz driver before being modulated onto a MZM biased in the quadrature point. To provide a basis for comparison a standard coherent transmitter based on an I/Q modulator was used to transmit BPSK and QPSK signals as well. The signals were transmitted b2b and over 20km of SSF fiber and attenuated before being detected. At the receiver side a single polarization (SP) 32 (class 20) and 64 GBaud (class 40) coherent receiver as well as a SOA and EDFA-filter PIN were used. The EDFA was a LNHPFA-23 from Pritel with a NF~5 dB and the SOA was the IPSAD1501 from Inphenix with a NF~8 dB. The filter was Gaussian with -3 dB bandwidth of 1.4 nm. For DD, a 42 GHz Thorlabs RXM42AF PIN/TIA was used. A micro-integrable tunable laser assembly (μTLA) was utilized as the ONU transmitter and used as the COH LO laser at the OLT receiver, which was set at +15.5 dBm. Both μTLAs were set close to λ=1550.2 nm. A polarization controller was used to align the signal polarization with that of the LO. A 64 GHz, 160 GS/s oscilloscope was used to capture the data for timing recovery and off-line processing in MatLab. All measurements were done in continuous mode

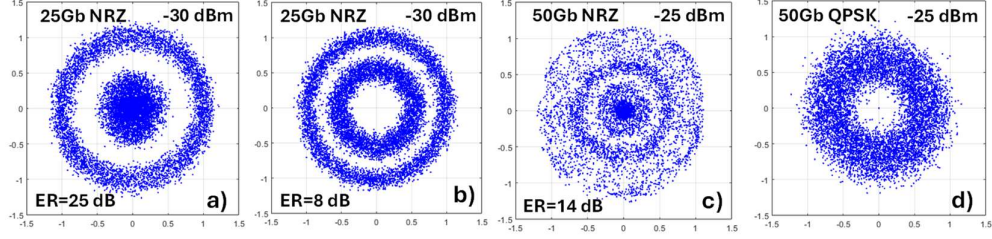


Fig. 2: Received constellations for a) and b) NRZ under different ER and c) with BW limited detection and d) for QPSK transmission all measured with a class 20 coherent receiver

(CM). Constant modulus algorithm (CMA) equalization and carrier and phase recovery was used for SP-BPSK and SP-QPSK detection. For NRZ signals using conventional CMA is challenging due to its zero level, so radius directed (ER dependent) decision-aided algorithms will be needed, see Figure 2. Fortunately, there is no need to recover the carrier frequency and phase when NRZ signals are received. Instead, the signal can be recovered by simply squaring and summing the detected I and Q signals, see Figure 1. Not needing carrier recovery for NRZ can be a major advantage in the fast acquisition of the signal under BM operation for the upstream. After this simple tap spaced FFE and DFE equalization was performed instead of CMA, like equalization for conventional DD based receivers. For full bandwidth signals 21 tap FFE equalization is used, while for bandwidth limited reception a 41 tap FFE + 3 tap DFE is used. The ER of the NRZ transmitted signal was varied by changing the Optical Modulation Amplitude (OMA) of the signal (difference in optical powers between the zero and one level), while keeping the average received power (P_{mean}) constant. To reach higher ER of 8.5 dB for 120 Gb/s NRZ the bias of the MZM was shifted towards the null point of the MZM as much as possible which reduced P_{mean} , see Figure 3b) (black curve). For the highest ER, the MZM was biased in the null and Optical Duobinary (ODB) was transmitted which enabled ER=30 dB at 50 Gb/s [3], see Figure 4b). Figure 2 shows the received constellations for NRZ with different extinction ratios and for QPSK transmission. From Figure 2a) and b) it can be observed

that a reduction in extinction ratio will lead to the zero level no longer being centered at zero which introduces an ER penalty. Figure 2c) shows the bandwidth limited reception of 50 Gb/s NRZ using a 32 Gbaud (class 20) coherent receiver. It can be noted that a 3-level signal (electrical duobinary) is received, which is equalized back to NRZ using DFE based equalization [2-3].

Modelled Extinction Ratio Penalties

In the 50G PON standard [7] the sensitivity is specified in OMA and the relationship between OMA, ER and P_{mean} is defined in appendix I of [7] with the extinction ratio penalty (δ) given by [5]:

$$\delta_{\text{DD (PIN)}} \approx \frac{er + 1}{er - 1} \quad (1)$$

with $\text{ER (dB)} = 10 \cdot \log(er)$.

For an optical pre-amplified (OA) DD receiver, the extinction ratio penalty under high gain approximation, assuming signal-ASE beating noise only can be derived to be [5,6]:

$$\delta_{\text{DD (OA-PIN)}} \approx \left(\frac{er + 1}{er - 1} \right)^2 \quad (2)$$

For coherently detected NRZ signals the sensitivity scales with the difference in optical field amplitude instead of the difference in optical powers. The ER penalty of a coherently detected NRZ signal is related to how much extra signal power is needed to obtain the same Euclidian distance between the zero and one level and is under usual conditions (LO shot noise limited) given by:

$$\delta_{\text{COH (PIN)}} \approx \left(\frac{\sqrt{er + 1}}{\sqrt{er - 1}} \right)^2 \quad (3)$$

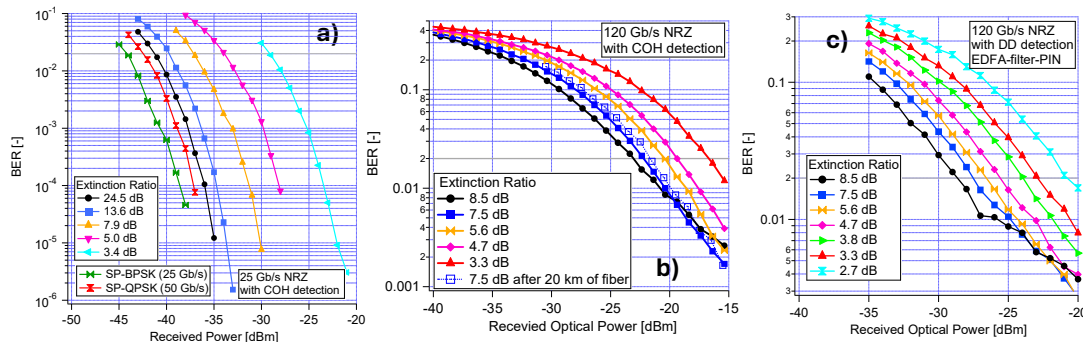


Fig. 3: BER Measurements of a) 25 Gb/s COH, b) 120 Gb/s COH and c) 120 Gb/s EDFA-PIN DD for different ER

BER Measurements for different ER

Figure 3a) depicts the BER measurements for COH detected 25 Gb/s NRZ. It can be noted that the receiver sensitivity degrades with reducing ER. A sensitivity of -41 dBm at BER=2e-2 was measured at ER=24.5 dB for COH detected 25G NRZ. As a comparison -44.2 dBm was obtained for BPSK and -42.5 dBm for QPSK at 25 Gbaud. The sensitivity of transmitted NRZ with $ER \rightarrow \infty$ should in theory closely match that of QPSK and will be 3 dB lower than BPSK transmission (see also Figure 4a). It's worth mentioning that since only one polarisation (SP) is received, a 3 dB sensitivity penalty will be incurred for a dual polarization (DP) implementation. Figure 3b) and c) shows the results for 120 Gb/s NRZ transmission for COH and DD detection. Due to transmitter bandwidth pre-compensation the maximum attainable ER is much lower for these baud-rates. A measured COH receiver sensitivity of -23.1 dBm and -28.7 dBm for DD for 120 Gb/s NRZ at BER=2e-2 for ER=8.5 dB was obtained. Since for COH detection we have full access to the phase and amplitude information in both polarizations, we performed CD compensation before timing recovery. We achieved <0.4 dB optical power penalty (OPP) for 120 Gb/s after 20 km of SMF fiber, see Figure 3b) (dotted line).

Measured ER penalties and Sensitivities

Figure 4 shows the measured sensitivities at BER=2e-2 of NRZ transmitted signals for COH and DD detection at different ER and baud rates. A good fit with calculated ER penalties at the different ER and baud-rates is obtained. Under the assumption of a minimum ER=6 dB for the ONU transmitter, same as specified in G.9804.3 [7] a receiver sensitivity of -20.8 dBm at 120 Gb/s and -23.7 dBm at 100 Gb/s for COH detection is obtained, see Figure 4a). This is much lower compared to what is typically reported for COH detection under BPSK and QPSK transmission [8] since NRZ transmission with ER=6 dB incurs a 10 dB penalty compared to BPSK [9] which can be calculated from (3) as well. This translates to a minimum average ONU transmit power of +8.5

dBm for 29 dB optical power budget assuming 0.3 dB CD penalty for 120 Gb/s. For the DD receiver a sensitivity of -26.9 dBm at 120 Gb/s and -30 dBm at 100 Gb/s for EDFA-filter-PIN and -25.6 dBm at 120 Gb/s for SOA-filter-PIN at BER=2e-2 is obtained for the same IM transmitter. The reduced sensitivity for SOA compared to EDFA is due to the larger NF of the SOA. It can be observed that OA-DD based receivers can outperform COH receivers under NRZ transmission, even for $ER \rightarrow \infty$ due to practical implementation penalties (e.g. excess loss) of integrated COH receivers [8,10,11]. Assuming a maximum 3.5 dB CD penalty [4] for DD a minimum average launch power of +7 dBm would be needed for a SOA-PIN implementation for 29 dB power budget, similar to COH detection. It should be noted that the sensitivity difference between 25 Gb/s and 50 Gb/s for DD in our case is small, as can be seen from Figure 4b) since the same 42 GHz bandwidth PIN was used for both rates. Also, the relatively large penalty between 100 Gb/s and 120 Gb/s is primarily caused by bandwidth limitations at the transmitter and can be improved upon.

Conclusions

The sensitivity of next generation 120 Gb/s IM upstream PON solutions as a function of ER has been studied. It was shown that the ER penalty depends on the receiver implementation and can become significant at low ER greatly affecting receiver sensitivities. Modelled ER penalties show good fit with measurements so Eq. (2) and Eq. (3) could be adopted for OA-DD and COH based receivers in future VHSP standards. COH and DD detection under NRZ transmission with ER=6 dB have similar launch power requirements. The benefit of COH detection is that the CD can be fully compensated for enabling C-band operation for ease of GPON co-existence. It was also shown that class 40 (64 Gbaud) coherent receivers can be used for 120 Gb/s NRZ reception in combination with low complex DSP, greatly reducing cost and ease of operation for upstream under BM operation.

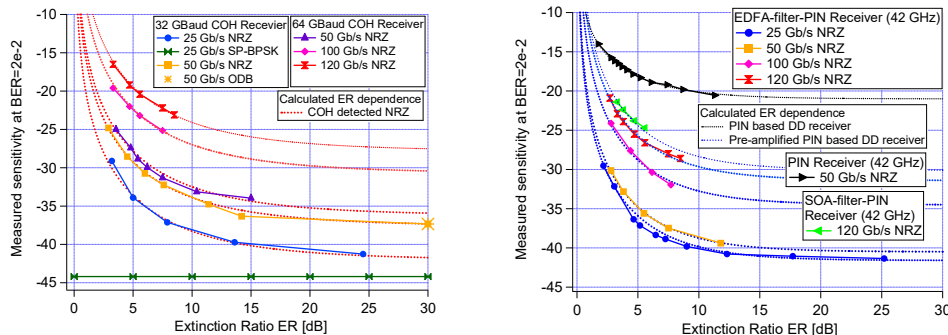


Fig. 4: Measured Extinction-Ratio penalty for a) COH and b) DD detection of transmitted NRZ signals

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